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# A ARCHITECTURE OPTIMIZATIONS

Many of our prompts require a concise summary of the agent, shorthanded as `[Agent's Summary Description]` in prompts above. In our implementation, this summary comprises agents' identity information (e.g., name, age, personality), as well as a description of their main motivational drivers and statements that describes their current occupation and self-assessment. Since this information is frequently used in many prompts, we synthesize it at regular intervals and access that synthesis as a cache.

To achieve this, we perform a retrieval on the query "[name]'s core characteristics." We then summarize the descriptors in the retrieved records by prompting the language model. For example:

```
How would one describe Eddy Lin's core characteristics
given the following statements?
- Eddy is a student at the Oak Hill College
studying music theory and composition
- Eddy is working on a new music composition [...]
```

This result: Eddy Lin is a student at Oak Hill College studying music theory and composition. He loves to explore different musical styles and is always looking for ways to expand his knowledge." We follow the same process in parallel on the queries "[name]'s current daily occupation" and "[name's] feeling about his recent progress in life". The agent's name, age, and traits are concatenated with these three summaries as the cached summary.

Many other optimizations are possible. Currently, because plans are likely to change from the agent's initial version, we only generate the high-level plan in advance and then recursively decompose the near future into the moment-to-moment action plan just in time. Another optimization might be to batch dialogue generation as a joint prompt rather than iterating back and forth between the agents, and re-planning could be architected to only invalidate and update parts of plans that strictly require adjustment. Our current architecture runs sequentially in roughly real-time game time (where one second real time is one minute game time), but it may be parallelized such that each agent runs in parallel.

# B AGENT INTERVIEW QUESTIONS

To evaluate generative agents, we conducted individual interviews by asking 25 questions that explore their abilities in five key areas: 1) self-knowledge, 2) memory, 3) planning, 4) reactions, and 5) reflections. Below, we have listed the interview questions utilized in our evaluation study and included a sample of responses from one of our simulated agents, Klaus Mueller.

## B.1 Self-Knowledge

The questions on agents' self-knowledge probe the agents' ability to retrieve relevant information about themselves and demonstrate coherence and consistency in their character.

- *Give an introduction of yourself.*
  Hello, my name is Klaus Mueller. I'm 20 years old and a student at Oak Hill College, studying sociology. I'm passionate about social justice and exploring different perspectives. Currently, I'm researching the effects of gentrification in low-income communities for a research paper.
- *What's your occupation?*
  I am a student studying sociology at Oak Hill College. I am currently writing a research paper on the effects of gentrification in low-income communities.
- *What is your interest?*
  I'm interested in writing and research. I'm writing a research paper on the effects of gentrification in low-income communities, and I'm passionate about exploring different perspectives and analyzing different points of view.
- *Who do you live with?*
  I live on my own.
- *Describe your typical weekday schedule in broad strokes.*
  My typical weekday starts with me waking up at 7:00 am and preparing for the day ahead. After that, I usually go to the library to work on my research paper, taking breaks for lunch at Hobbs Cafe and a walk in the park. In the evening, I often return to the library until around 6:00 pm.

## B.2 Memory

The questions on agents' memory probe the agents' ability to remember their past experiences and retrieve the relevant pieces of memory. The names in the square brackets of the first and last questions are randomly chosen from the names of other agents with whom the subject agent has interacted.

- *Who is [Wolfgang Schulz]?*
  Wolfgang Schulz is a person I met recently who has been helping me with my research paper on the effects of gentrification on low-income communities. We had several conversations about it, he gave me some great resources, and he even invited me to join him and another friend for lunch. He seems really curious about the issue and has a good grasp on it too. I found that really interesting.
- *Who is Kane Martinez?*
  I'm sorry, I'm not sure who Kane Martinez is. Could you tell me more about them?
- *Who is running for the election?*
  I've heard that Sam Moore is considering running for local mayor.
- *Was there a Valentine's day party?*
  Yes, Isabella Rodriguez organized a Valentine's Day party at Hobbs Cafe.
- *Who is [Ayesha Khan]?*
  Ayesha Khan is a fellow student at Oak Hill College. She is doing her senior thesis on the use of language in Shakespeare's plays. I am looking forward to speaking with her and exchanging tips with her, along with Wolfgang and Maria Lopez, who are both doing research.